

Radial-Based Oversampling for Multiclass Imbalanced Data Classification

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Abstract—Learning from imbalanced data is among the most popular topics in the contemporary machine learning. However, the vast majority of attention in this field is given to binary problems, while their much more difficult multiclass counterparts are relatively unexplored. Handling data sets with multiple skewed classes poses various challenges and calls for a better understanding of the relationship among classes. In this paper, we propose multiclass radial-based oversampling (MC-RBO), a novel data-sampling algorithm dedicated to multiclass problems. The main novelty of our method lies in using potential functions for generating artificial instances. We take into account information coming from all of the classes, contrary to existing multiclass oversampling approaches that use only minority class characteristics. The process of artificial instance generation is guided by exploring areas where the value of the mutual class distribution is very small. This way, we ensure a smart oversampling procedure that can cope with difficult data distributions and alleviate the shortcomings of existing methods. The usefulness of the MC-RBO algorithm is evaluated on the basis of extensive experimental study and backed-up with a thorough statistical analysis. Obtained results show that by taking into account information coming from all of the classes and conducting a smart oversampling, we can significantly improve the process of learning from multiclass imbalanced data.

Index Terms—Imbalanced data, machine learning, multiclass imbalance, oversampling.

I. INTRODUCTION

LEARNING from imbalanced data has become the focus of intense research due to its compliance with real-world pattern classification tasks, where the most important or interesting classes are usually strongly underrepresented [1], [2]. Most studies focus on the binary problem, where the well-represented class is known as the majority class and the underrepresented class is known as the minority one. In such a case, a learning difficulty usually occurs, as traditional classification algorithms tend to be biased toward the majority class [2]. Nevertheless, in the case of the imbalanced

data classification, the disproportion between different classes is not the sole issue of learning difficulties. One may easily come up with an example where the instance distributions from different classes are well separated despite a high imbalance ratio (IR). In such a case, proposing an efficient classifier is not challenging. Unfortunately, instances from the minority class often form small clusters of an unknown structure that are scattered [3]. The additional complication comes from the fact that during learning, the number of learning examples from the minority class may be not sufficient for the learning algorithm to acquire the appropriate generalization level, which in effect may lead to an overfitted model [4].

The mentioned challenges in learning from imbalanced data are a focus of intense research [5]–[7], yet mostly in the context of binary problems. However, in a vast number of real-life problems, we are faced with multiclass data sets that are characterized by skewed distributions. While for binary problems, relationships between the classes are easy to be defined, and in multiclass problems, the mutual relationships among classes are much more complex [8]. Popular methods dedicated to two-class problems cannot be directly applied to their multiclass counterparts, as they cannot model the relationships among classes and inherited learning difficulties, such as multiple classes overlapping or borderline instances being located among more than two decision regions. One may decompose a given problem into a set of binary subproblems, yet this is an oversimplification that loses a valuable perspective on nonpairwise data characteristics [1], [9].

Oversampling algorithms have gained a significant popularity in the binary imbalanced domain [9], and one can see several recent proposals of extending them into the multiclass area [10]. Oversampling algorithms are especially attractive to multiclass domains as they work in a preprocessing manner. Therefore, a data set balanced with artificial instances can be given to any classifier. This is easier, more flexible, and versatile than careful modification of a single existing classified to become skew-insensitive [9]. However, existing algorithms dedicated to both binary and multiclass data sets suffer from similar limitations [1]. Their main drawback lies in taking into account only information coming from a minority class (in multiclass domain one of the minority classes at a time). Such an approach completely ignores the information coming from other classes and leads to often imperfect placement of artificial instances. In addition, the overlapping among classes may be increased or the original distribution of a given class may be shifted [11]. Furthermore, most of the existing oversampling algorithms introduce the artificial instances based on the selected nearest neighbors (NNs). This is very

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sensitive to atypical data distributions, overlapping classes, and small disjuncts. Such data-level difficulties become even more prominent in a multiclass setting, thus calling for the development of more guided solutions [10].

To alleviate the identified drawbacks of the existing algorithms, this paper introduces a novel algorithm named multiclass radial-based oversampling (MC-RBO). It is developed specifically to handle imbalanced problems with multiple classes and embedded data-level difficulties, i.e., atypical data distributions, overlapping classes, and small disjuncts. The main strength of MC-RBO lies in utilizing potential functions calculated using every instance from each class. Therefore, we make an important step toward a new view on the oversampling scheme, by showing that utilizing information coming from all of the classes is highly beneficial. We use the calculated potential values to place artificially generated instances in a guided manner that will not lead to any changes in original class characteristics. Our methodology steps away from using NNs for instance generation, thus further reducing the shortcoming of existing methods. This way, we achieve a guided and smart oversampling in the presence of multiple classes.

To summarize, the contributions of this paper are as follows.

- 1) Proposition of the MC-RBO algorithm, which allows for intelligent data oversampling that exploits local data characteristics of each class.
- 2) Utilization of information coming from all of the classes during the artificial instance generation procedure that offers better placement of new instances and more targeted empowering of minority classes.
- 3) New approach for generating artificial instances that do not use NNs of each minority instance for instance imputation, thus making MC-RBO robust to atypical data distributions.
- 4) Detailed analysis of computational complexity of our method, showcasing its reliable tradeoff between pre-processing time and obtained improvements in handling imbalanced data.
- 5) Experimental evaluation of the proposed approach on the basis of diverse benchmark data sets and a detailed comparison with the state-of-art approaches.

This paper is organized as follows. Section II discusses in detail the problem of learning from imbalanced data and emphasizes the unique characteristics of multiclass problems. Section III discusses the shortcomings of existing oversampling approaches and introduces MC-RBO in detail, while Section IV depicts the conducted experimental study. Section V concludes this paper and offers insight into future directions in this field.

II. LEARNING FROM IMBALANCED DATA

Imbalanced class distributions can pose a significant learning difficulty to most of the classifiers [1], [2]. The underlying disproportion between majority and minority instances, known as IR, may negatively impact any training process that is guided by accuracy or uniform loss functions. By assuming an equal importance of all training instances, favoring the

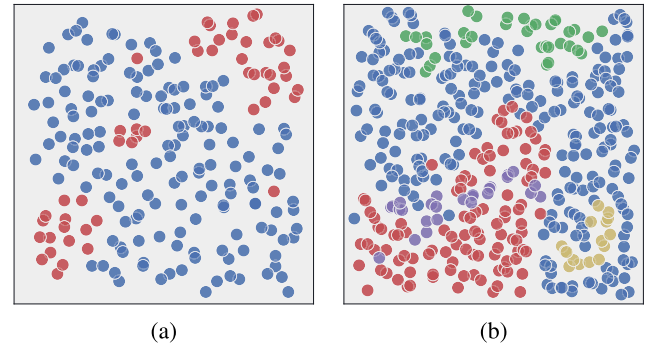


Fig. 1. Different imbalanced scenarios. (a) Binary imbalanced data set. (b) Multiclass imbalanced data set.

majority class will lead to the best accuracy (or the lowest loss). Such a bias toward the majority class will impair the generalization capabilities over the minority class, thus leading to an inefficient learning system. The class imbalance has been mainly studied in the context of binary problems, yet it may also be present in a much more challenging multiclass setting (see Fig. 1).

There are two major approaches to the class imbalance problem as follows.

Data-Level Solutions: These methods rebalance the training set, in order to make it suitable for any standard classifier. This is achieved by either reducing the number of majority class instances (undersampling) or increasing the number of minority class instances (oversampling). As random techniques do not offer any control over the process of rebalancing, guided solutions that can preserve the minority class properties have been studied. The most popular solution based on this philosophy is Synthetic Minority Oversampling Technique (SMOTE), which is, nowadays, considered as a cornerstone for the majority of the proposed oversampling algorithms [12], [13]. Unfortunately, SMOTE may lead to changes in the characteristic of the minority class and in result to *overfitting* the classifier because it assumes that instances from minority class form a homogeneous area (or multiple homogeneous clusters) [9]. We will discuss the limitations of SMOTE-based methods in Section III-A. Several modifications of SMOTE have been proposed, which can identify the instances that will become a base to generate new artificial ones in a more guided fashion such as *borderlines*SMOTE [14]. It generates new instances from the minority class close to the decision border. *Safe-level* SMOTE [6] and LN-SMOTE [15] reduce the probability of generating synthetic instances of the minority class in areas where the predominant objects are that of the majority class. Koziarski *et al.* proposed two novel solutions to this problem: RBO [16] and combined cleaning and resampling [17] that enforce instances from the majority class to be relocated from the areas where the minority-class instances are present. Sharma *et al.* [18] proposed sampling with the majority, an efficient approach for combining information from both minority and majority classes to insert new instances in a meaningful manner.

Algorithm-Level Solutions: These methods identify what specific mechanism in the classifier training routine fails in

the context of imbalanced data. Then, it is being replaced with a skew-insensitive modification, effectively rendering a given classifier robust to uneven class distributions. Popular examples include splitting criteria in decision trees [19], kernel functions [20], or modifying the underlying loss function to make it cost-sensitive [21]. Algorithm-level modifications require in-depth knowledge about a given classifier and are restricted only to the selected model [22] but are often reported to work more efficiently than sampling approaches [9].

Mentioned solutions are commonly hybridized with ensemble learning to harness the improved accuracy and robustness of combining classifiers [23]. Approaches that combine sampling or cost-sensitive learning with bagging [24] and boosting [25] are the most popular ones.

Recent studies point to the fact that IR is not the sole source of learning difficulties in imbalanced data [1], [9]. Imbalanced data sets are often accompanied by a plethora of difficulties embedded in the nature of data. A small sample size of the minority class may cause a lack of representativeness of training data, thus rendering all sampling methods ineffective. Borderline and overlapping instances pose an additional challenge to any classifier, and they will limit the capabilities of reaching a good tradeoff between performance on minority and majority classes. Noisy instances and outliers may mislead a classifier and cause overfitting.

A. Multiclass Imbalanced Problems

Multiclass imbalanced problems are significantly more challenging than their binary counterparts, yet not as much attention has been given to them [1], [8]. We need to deal with complex relationships among classes, as the notion of minority and majority classes is not so distinct anymore. One of the earliest proposals for multiclass imbalance taxonomy postulated “multiminority” (a single majority class and multiple minority ones) and “multimajority” (multiple majority classes and a single minority) problems [8]. However, this taxonomy is very restrictive, as in real-world scenarios, one will never deal with such a clearly defined case. On the contrary, a single class may be a minority for some classes, while in rough balance with some, and majority to others [1]. This causes a significant learning difficulty, as one can easily lose performance on some classes while gaining on others.

Taxonomy for methods dedicated to multiclass imbalance differs from the binary one. While the division between data and algorithm-level solutions still holds, algorithms differ in the way they approach the multiclass nature of the problem:

1) *Binarization Solutions*: These methods decompose the M -class problem into $M(M - 1)/2$ [one-versus-one (OVO)] or M [one-versus-all (OVA)] binary subproblems [26]. Each decomposed task is handled using a selected algorithm for binary imbalanced data and dedicated combination algorithms reconstruct the multiclass decision from a set of binary ones. The advantage of this approach lies in the ease of utilizing existing algorithms [27]. This, however, comes at a significant price, as binarization breaks the relationships among multiple classes and oversimplifies them into pairwise relationships. Therefore, we lose a significant amount

of valuable information on the characteristics of different classes. In addition, the number of OVO classifiers grows quadratically with the number of classes and hence the efficient reconstruction of the multiclass output may be hindered. For OVA, we combine $M - 1$ classes into the negative class, effectively reinforcing the class imbalance, instead of alleviating it. This makes usage of canonical OVA decomposition ill-advised when dealing with problems characterized by a high number of classes with a significant IR. Recent works in this domain suggest the efficacy of using ensemble methods with OVO decomposition [28], augmenting it with cost-sensitive learning [29] or applying dedicated classifier combination methods as error correcting output codes [30].

2) *Ad Hoc Solutions*: These methods treat the problem natively as a multiclass one and try to exploit complex and nonpairwise relationships among classes. They require developing new algorithms, as the binary ones cannot be directly applied. The cost of more complex algorithm design and development is overcompensated with frequent excellent performance, as *ad hoc* algorithms are closer into grasping core problems in multiclass imbalanced data. Most popular data-level approaches include modifications of SMOTE algorithm to take into account more than two classes [10], [31], [32], combination with feature selection [33], [34], and alternative methods for instance generation by using Mahalanobis distance [35], [36]. Algorithm-level solutions include decision tree adaptations [37], cost-sensitive matrix learning [38], and ensemble solutions utilizing bagging [24], [39] and boosting [8], [40].

It is important to note that the difficulties embedded in the nature of imbalanced data are even more prevalent in multiclass scenarios. Class overlap or borderline instances may come from more than two classes, making such uncertainty regions especially difficult for any classifier. One must carefully consider small disjuncts and outliers, as they may influence the decision boundaries for several classes at once. Finally, any data-level instance generation or removal must be done by a careful analysis on how action on a single class influences different types of instances in remaining classes. This allows us to conclude that there is a need to develop new and dedicated approaches to multiclass imbalanced data that will take into account various associated learning difficulties.

III. RADIAL-BASED OVERSAMPLING FOR MULTICLASS PROBLEMS

To address the problems associated with the multiclass imbalanced data, we propose a novel oversampling strategy, MC-RBO. It was inspired by the limitations of the most prevalent approaches to the imbalanced data oversampling, neighborhood-based strategies, and designed to alleviate them. In the remainder of this section, we describe the proposed method. We begin with a discussion on the possible shortcomings of the neighborhood-based oversampling strategies, with particular attention given to SMOTE. Next, we propose an alternative approach of choosing the regions, in which synthetic samples should be generated, which is estimating the *class potential*. Afterward, we describe how the class potential

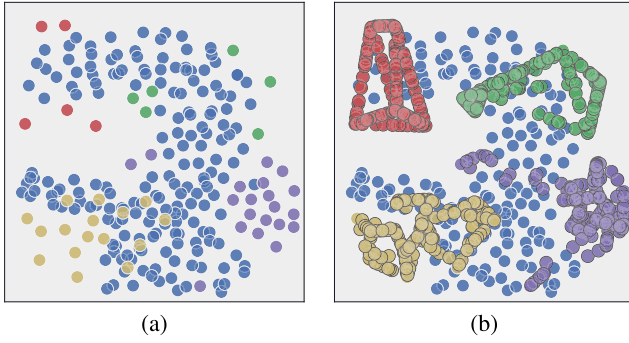


Fig. 2. Limitations of neighborhood-based oversampling strategies in the multiclass setting illustrated with an example of STATIC-SMOTE. Neighborhood-based strategies are susceptible to disjoint data distributions and presence of noise, and outliers. (a) Initial data set. (b) Data set after balancing with STATIC-SMOTE.

can be used to guide the oversampling procedure in the binary setting, in the form of binary radial-based oversampling (B-RBO) algorithm. Finally, we extend the B-RBO algorithm to the multiclass setting.

A. Limitations of Neighborhood-Based Oversampling Strategies

The critical decision in the design of the imbalanced data oversampling algorithms is the choice of the *regions of interest*, the areas, in which we consider placing the newly generated minority observations. For instance, in the case of random oversampling (ROS), the existing minority observations are being duplicated—the regions of interest, in this case, are the points, in which the minority observations are located. On the other hand, in the case of SMOTE, synthetic minority observations are being placed between the nearest minority neighbors—the regions of interest are located between these objects. While too centralized regions of interest, such as the ones produced by ROS, can lead to overfitting [5], placing the regions of interest over the areas with a high density of majority observations can negatively impact the detection of majority classes. We argue that the neighborhood-based approaches, such as SMOTE and its derivatives, are an example of methods using too intrusive regions of interest when applied to a difficult data sets, characterized by the presence of disjoint data distributions, noise, and outliers. The main reason causing this behavior is the fact that this family of methods uses solely the distribution of the minority class, neglecting the placement of majority observations. We illustrate this point in Fig. 2, in which we used STATIC-SMOTE to oversample a difficult data set. Produced synthetic observations often overlap the clusters of majority observations, neglecting their position. Classifier trained on such oversampled data set would likely produce decision borders biased toward the minority classes, even though it might not be justified.

The severity of the issue can be partially mitigated by reducing the number of NNs used by the SMOTE. However, the problem of singular outliers would withstand. Furthermore, it is not clear what the number of neighbors should be chosen, or how this number could be adjusted locally

instead of a single, global value for the whole data set. Finally, in principle, a cleaning procedure, such as Tomek links [41] or edited NN rule [42], could be used to remove the synthetic minority observations overlapping the majority clusters. Nevertheless, achieving balanced class distributions with such approach could be complicated, since even if we generate a more significant number of synthetic observations during the oversampling, and in degenerate cases, they could still all be removed during the cleaning step.

B. Class Potential Estimation With Radial-Basis Functions

As an alternative to the regions of interest used by the neighborhood-based strategies, we propose an approach using the information about the mutual class distribution. The method will be based on the notion of *potential*. Intuitively, for a collection of observations $\mathcal{X}$, a potential in any point in space x will represent the cumulative proximity of that point to said observations. In other words, a point x will be characterized by a higher potential concerning a given collection of observations, if it is located nearby a large number of observations from that collection. To model this behavior, we will attach a radial basis function (RBF) to every observation from the collection. RBF is a real-valued function, the value of which depends on its distance from the point of origin.

Using the concept of Gaussian RBF, we define the potential function as follows: given a collection of observations $\mathcal{X}$, parameter γ representing the spread of a single RBF, a point in space x , and using the $L1$ -norm as a distance measure, the potential function is defined as

$$\Phi(x, \mathcal{X}, \gamma) = \sum_{i=1}^{|\mathcal{X}|} e^{-\left(\frac{\|\mathcal{X}_i - x\|_1}{\gamma}\right)^2}. \quad (1)$$

We will further denote the subcollection of observations $\mathcal{X}$ belonging to class c by $\mathcal{X}^{(c)}$. Of particular interest will be the *class potential*, the potential computed on the collection of observations belonging to a specific class $\mathcal{X}^{(c)}$, and the *mutual class potential*, the difference between the potential of two different classes. While discussing the potentials, in the binary setting, we will assume the following convention: high mutual majority class potential indicates the situation, in which the class potential for the majority class is higher than the potential of the minority class, and high mutual minority class potential indicates the opposite. An example of the mutual class potentials computed for a simple multiclass data set was presented in Fig. 3. As can be seen, the choice of the γ parameter affects the influence of more distant observations on the potential in a given point: using higher values of γ leads to smoother potentials, whereas the lower values produce potentials centralized around the existing observations. In all cases, the potential function for a given class takes the highest values near and inside the clusters of observations of that class. On the other hand, mutual class potential has the lowest absolute values in the areas of low certainty, in which class distributions overlap.

In the context of imbalanced data oversampling, potential representation carries more information than the regions of

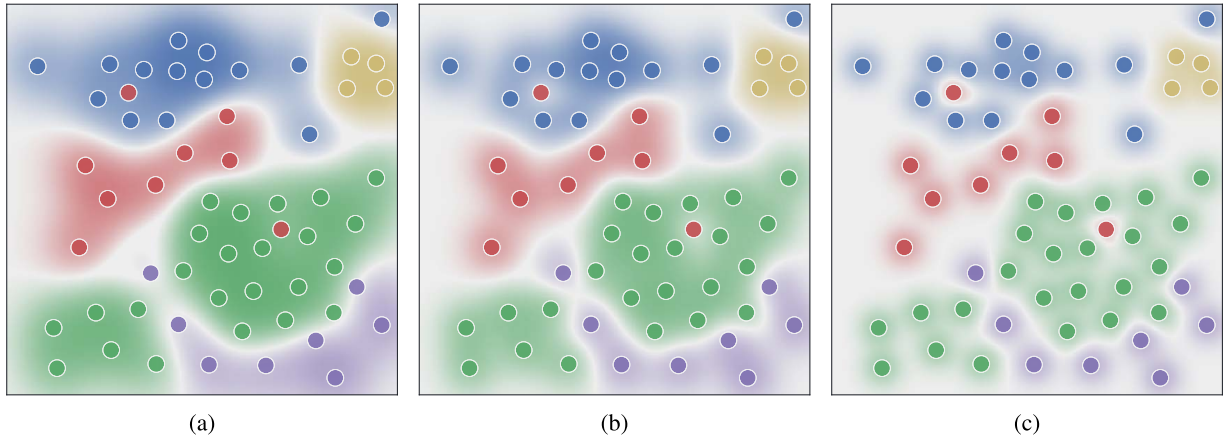


Fig. 3. Visualization of class potentials for different values of γ parameter. (a) $\gamma = 0.100$. (b) $\gamma = 0.075$. (c) $\gamma = 0.050$.

interest produced by neighborhood-based methods. While the latter is located inside the regions connecting neighboring minority observations and do not accommodate the information about the majority distribution, the potential representation uses both.

C. Binary Radial-Based Oversampling

In the binary setting, class potential can be used to guide the oversampling procedure in various ways, and based on the cost, we assign to either precision or recall. Depending on the type of problem at hand, one could either place the synthetic observations in the areas of high mutual minority class potential, which should have least impact on the correct detection of the majority class, or high mutual majority class potential, which should move the decision border in favor of minority class. The variant we consider in this paper is based on placing the synthetic minority observations in the areas of mutual class potential close to zero, which represent high uncertainty we assign to the belongingness of a given point to either class. It is a compromise between either the high mutual majority or minority class potential. Based on the results of the preliminary analysis conducted in the binary setting, partially described in [16], it also led to empirically best results.

In addition to the optimization criterion, an optimization algorithm is necessary to find suitable regions and actually produce the synthetic observations. In this paper, we used a modification of a simple hill climbing algorithm. Starting in the position of a randomly chosen, existing minority observation, we translated it in a random direction. If the absolute value of the mutual class potential was lower after the translation, it was kept. To reduce the computational cost of the algorithm, we considered the possibility of potential approximation: instead of computing the value of potential based on all of the existing observations, we used only k NNs of the initial minority observation. The whole procedure was repeated for a set number of iterations. The oversampling process lasted until the class distributions were completely balanced. The pseudocode of the described algorithm was presented in Algorithm 1.

Algorithm 1 B-RBO

- 1: **Input:** collections of majority observations $\mathcal{X}_{maj}$ and minority observations $\mathcal{X}_{min}$
 - 2: **Parameters:** spread of radial basis function γ , optimization *step*, number of *iterations* per synthetic observation, k nearest neighbors used for potential approximation
 - 3: **Output:** collection of synthetic minority observations S
 - 4:
 - 5: **function** B-RBO($\mathcal{X}_{maj}$, $\mathcal{X}_{min}$, γ , *step*, *iterations*, k):
 - 6: $S \leftarrow \emptyset$
 - 7: find k nearest neighbors from $\mathcal{X}_{maj} \cup \mathcal{X}_{min}$ for every observation in $\mathcal{X}_{min}$
 - 8: **while** $|\mathcal{X}_{min}| + |S| < |\mathcal{X}_{maj}|$ **do**
 - 9: $x^0 \leftarrow$ randomly chosen observation from $\mathcal{X}_{min}$
 - 10: $\mathcal{X}_{maj}^0, \mathcal{X}_{min}^0 \leftarrow$ collections of majority and minority observations belonging to k nearest neighbors of x^0
 - 11: $\Phi^0 \leftarrow \Phi(x^0, \mathcal{X}_{maj}^0, \gamma) - \Phi(x^0, \mathcal{X}_{min}^0, \gamma)$
 - 12: **for** $i \leftarrow 1$ **to** *iterations* **do**
 - 13: *direction* $\leftarrow$ randomly chosen standard basis vector in a m -dimensional Euclidean space, with m being the number of features
 - 14: *sign* $\leftarrow$ randomly chosen value from $\{-1, 1\}$
 - 15: $x' \leftarrow x^0 + \textit{direction} \times \textit{sign} \times \textit{step}$
 - 16: $\Phi' \leftarrow \Phi(x', \mathcal{X}_{maj}^0, \gamma) - \Phi(x', \mathcal{X}_{min}^0, \gamma)$
 - 17: **if** $|\Phi'| < |\Phi^0|$ **then**
 - 18: $x^0 \leftarrow x'$
 - 19: $\Phi^0 \leftarrow \Phi(x', \mathcal{X}_{maj}^0, \gamma) - \Phi(x', \mathcal{X}_{min}^0, \gamma)$
 - 20: **end if**
 - 21: **end for**
 - 22: $S \leftarrow S \cup \{x^0\}$
 - 23: **end while**
 - 24: **return** S
-

D. Multiclass Radial-Based Oversampling

To extend the B-RBO algorithm to the multiclass setting, we propose an iterative procedure, during which classes will be oversampled individually, one at the time. First of all, we sort the classes by the number of associated observations in the

descending order. We pick the majority class, the class with the highest number of observations. Afterward, for each of the classes except the majority class, we construct a collection of combined majority observations, consisting of a fraction of observations from each of the already considered class. Finally, we perform the oversampling with B-RBO algorithm, using the observations from the currently considered class as a minority, and the combined majority observations as the majority class. The pseudocode of the proposed algorithm was presented in Algorithm 2. Compared with an alternative strategy of adapting B-RBO to the multiclass setting, OVA class decomposition, the proposed method displays two advantages. First of all, it usually reduces the computational cost of the algorithm, since the collection of combined majority observations was often smaller than the collection of all observations in our experiments. Second, it assigns equal weight to every class in the collection of combined majority observations since each of them has an equal number of observations. This would not be the case in OVA decomposition, in which the classes with a higher number of observations could dominate the rest. Based on our observations, using the proposed class decomposition strategy usually also led to achieving a better performance during the classification.

Algorithm 2 MC-RBO

```

1: Input: collection of observations  $\mathcal{X}$ , with  $\mathcal{X}^{(c)}$  denoting a
   subcollection of observations belonging to class  $c$ 
2: Parameters: spread of radial basis function  $\gamma$ , optimization
   step, number of iterations per synthetic observation,
    $k$  nearest neighbors used for potential approximation
3: Output: collection  $\mathcal{S}$ , with  $\mathcal{S}_i$  denoting a collection of
   synthetic minority observations generated for  $i$ -th class
4:
5: function MC-RBO( $\mathcal{X}$ ,  $\gamma$ , step, iterations,  $k$ ):
6:    $\mathcal{S} \leftarrow \emptyset$ 
7:    $C \leftarrow$  collection of all classes, sorted by the number of
     associated observations in a descending order
8:   for  $i \leftarrow 1$  to  $|C|$  do
9:      $n_{classes} \leftarrow$  number of classes with higher number of
       observations than  $C_i$ 
10:    if  $n_{classes} = 0$  then
11:       $\mathcal{S}_i \leftarrow \emptyset$ 
12:    else
13:       $\mathcal{X}_{min} \leftarrow \mathcal{X}^{(C_i)}$ 
14:       $\mathcal{X}_{maj} \leftarrow \emptyset$ 
15:      for  $j \leftarrow 1$  to  $n_{classes}$  do
16:        add  $\lfloor \frac{|\mathcal{X}^{(C_j)}|}{n_{classes}} \rfloor$  randomly chosen observations from
           $\mathcal{X}^{(j)} \cup \mathcal{S}_j$  to  $\mathcal{X}_{maj}$ 
17:      end for
18:       $\mathcal{S}_i \leftarrow$  B-RBO( $\mathcal{X}_{maj}$ ,  $\mathcal{X}_{min}$ ,  $\gamma$ , step, iterations,  $k$ )
19:    end if
20:  end for
21: return  $\mathcal{S}$ 

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An illustration of the behavior of MC-RBO compared with SMOTE is presented in Fig. 4. As can be seen, contrary to SMOTE, MC-RBO is resilient to the presence of outliers in

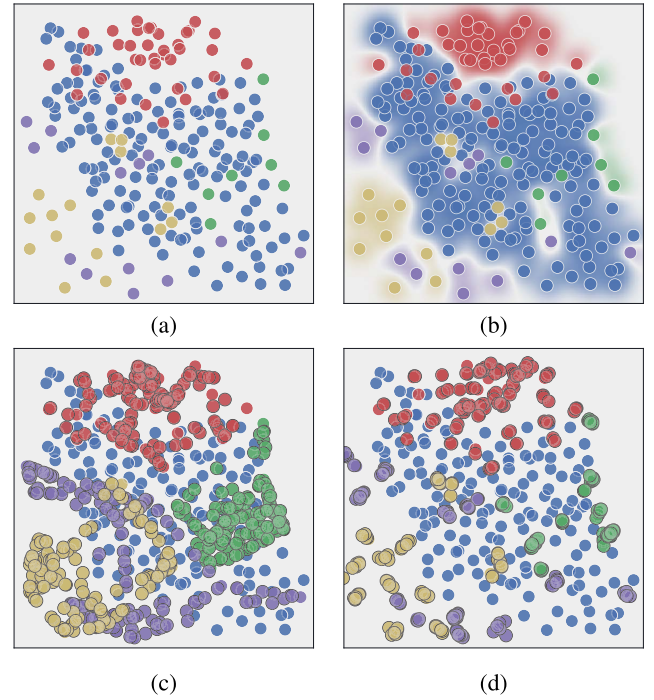


Fig. 4. Example of an imbalanced data set, calculated class potentials, and comparison of STATIC-SMOTE and MC-RBO oversampling. MC-RBO produces more conservative synthetic observations, which are much less likely to overlap the distributions of other classes. (a) Initial data set. (b) Class potentials. (c) Data set after balancing with STATIC-SMOTE. (d) Data set after balancing with MC-RBO.

a sense that generated synthetic minority observations do not overlap the clusters of other classes. Instead, in the case of outliers, they are located in a close proximity to the existing observations. On the other hand, when not affected by the distributions of other classes, they are located further away from the original observations. It should be noted that in the presented example, in either case, oversampled synthetic observations are located relatively close to the original observations. This spread can be adjusted by increasing the number of iterations.

Our method is published as an open-source code in Python and it is publicly available for download.¹

E. Computational Complexity Analysis

Let us define the total number of observations by n , the number of majority and minority observations in the binary case by n_{maj} and n_{min} , respectively, the number of features by m , the number of algorithms iterations per synthetic observation by i , the number of NNs used to approximate the potential function by k , and the number of classes in the multiclass case by c . Let us first consider the worst case complexity of the binary variant of RBO without the potential approximation. A single calculation of the potential in any given point with respect to $\mathcal{X}$, as defined in (1) requires $|\mathcal{X}|$ distance calculations, each with a complexity of $\mathcal{O}(m)$, a single summation and a single RBF calculation. The mutual class potential requires

¹<https://github.com/michalkoziarski/MultiClassRBO>

the calculation of the potential for both majority and minority classes, as well as a single subtraction and two calculations of the absolute value, for a total complexity of $\mathcal{O}(mn)$. The binary variant of RBO without the potential approximation can be simplified, without impacting the computational complexity to i mutual class potential calculations per generated synthetic observation, with $n_{\text{maj}} - n_{\text{min}}$ synthetic observations generated. In the worst case $n_{\text{min}} = 1$ and $n_{\text{max}} = n - 1$, which results in the worst case complexity of the algorithm equal to $\mathcal{O}(imn^2)$.

In comparison, using potential approximation reduces the computational complexity of the generation stage to $\mathcal{O}(ikmn)$ since the number of objects used during a single mutual class potential calculation is reduced from n to k . However, an additional step, in which k NNs must be found for every minority observation, is required. This can be divided into the following: 1) calculating the distance matrix between the minority observations, with the complexity of $\mathcal{O}(mn_{\text{min}}^2)$; 2) calculating the distances between every majority and minority observation, with the complexity of $\mathcal{O}(mn_{\text{maj}}n_{\text{min}})$; and 3) sorting an n -dimensional vector of distances for every minority observation, with the complexity of $\mathcal{O}(n_{\text{min}}n \log n)$. Simplifying, the worst case complexity of the k NN search is equal to $\mathcal{O}(mn^2 + n^2 \log n)$, and the final worst case complexity of the binary RBO using potential approximation is equal to $\mathcal{O}(mn^2 + n^2 \log n + ikmn)$.

Finally, the worst case for the multiclass setting occurs when $c - 1$ classes consist of 1 observation each, and a single class consists of $n - c + 1$ observations, or in other words when a single majority class consists of the maximum possible number of observations. In this case, the binary RBO is called $c - 1$ times, and every time $n - c$ synthetic observations are generated. Therefore, the worst case complexity of the MC-RBO algorithm is equal to $\mathcal{O}(c(mn^2 + n^2 \log n + ikmn))$.

IV. EXPERIMENTAL STUDY

In this section, we will describe the details of a conducted experimental study that allows verifying the usefulness of the proposed MC-RBO algorithm. The experiments were designed to answer the following questions.

- 1) What is the relative performance of MC-RBO when compared to other *ad hoc* multiclass solutions coming from both data-level and algorithm-level solutions?
- 2) How does MC-RBO compare with using standard binary sampling methods combined with multiclass decomposition approaches?
- 3) Is MC-RBO flexible enough to work efficiently with various types of classifiers?
- 4) What is the impact of the way we select instances for other classes when calculating potential functions in MC-RBO?

The following sections present information on used multiclass imbalanced benchmarks (Section IV-A), set-up of the experimental study (Section IV-B), as well as discussion of obtained results (Section IV-C).

TABLE I
DETAILS OF MULTICLASS IMBALANCED BENCHMARKS USED IN THE EXPERIMENTS

Dataset	#Instances	#Features	#Classes	IR	Class distribution
Automobile	150	25	6	16.00	3/20/48/46/29/13
Balance	625	4	3	5.88	288/49/288
Car	1728	6	4	18.61	65/69/384/1210
Cleveland	297	13	5	12.62	164/55/36/35/13
Contraceptive	1473	9	3	1.89	629/333/511
Dermatology	358	33	6	5.55	111/60/71/48/48/20
Ecoli	336	7	8	71.50	143/77/22/35/20/5/52
Flare	1066	11	6	7.70	331/239/211/147/95/43
Glass	214	9	6	8.44	70/76/171/39/29
Hayes-Roth	160	4	3	2.10	160/65/64/31
Led7digit	500	7	10	1.54	45/37/51/57/52/47/57/53/49
Lymphography	148	18	4	40.50	2/81/61/4
New-thyroid	215	5	3	5.00	150/35/30
Page-blocks	5472	10	5	175.46	4913/329/28/87/115
Thyroid	7200	21	3	40.16	166/368/6666
Vehicle	846	18	4	1.17	199/212/17/218
Wine	178	13	3	1.48	59/71/48
Winequality-red	1599	11	6	68.10	10/53/681/638/199/18
Yeast	1484	8	10	92.60	244/429/463/44/51/163/35/30/20/5
Zoo	101	16	7	10.25	41/13/10/20/8/5/4

A. Multiclass Imbalanced Data Sets

For this paper, we have selected diverse 20 multiclass imbalanced data sets as benchmarks. They can be downloaded from the KEEL repository [43]. Their details are provided in Table I.

We have selected these benchmarks on the basis of our previous study on the instance-level difficulties in multiclass data sets [10]. Our analysis showed that they have embedded many challenging characteristics, such as small disjuncts, frequent borderline and noisy instances, and class overlapping. A detailed discussion can be found in [10].

B. Setup

1) *Reference Methods*: We aimed at comparing MC-RBO with state-of-the-art approaches to multiclass imbalanced data. These include three groups of methods: 1) multiclass sampling approaches; 2) multiclass algorithm-level approaches; and 3) binary decomposition approaches. The first group consists of using SMOTE with round-robin approach for simple class balancing (SMOTE-all), STATIC-SMOTE (S-SMOTE) [31], Mahalanobis distance oversampling (MBO) [35], and k -NN-based synthetic minority oversampling algorithm to combat multiclass imbalance problems (SMOM) [32]. The second group consists of multiclass Hellinger decision tree (HTree) [37], near-Bayesian support vector machines (NB-SVMs) [44], boosting-based AdaBoost.NC ensemble (AdaB.NC) [8], and bagging-based SMOTEBagging ensemble (SMOTEBag) [45]. The third group uses OVO decomposition with a weighted voting combination for reconstructing the multiclass task and four distinct sampling approaches: random undersampling (RUS), SMOTE [5], Borderline SMOTE 1 (BSMOTE) [14], and adaptive synthetic sampling approach (ADASYN) [46]. Detailed parameter ranges for examined algorithms are given in Table II.

2) *Base Classifiers*: In order to verify the flexibility of MC-RBO, we have examined its effects on four different base classifiers: decision trees (C5.0 model), neural networks (multilayer perceptron with backpropagation, MLP), lazy learners (k -NNs, k -NN), and probabilistic classifiers [Naïve Bayes (NB)]. Detailed parameter ranges for used classifiers are given in Table II.

TABLE II
PARAMETERS FOR THE USED CLASSIFIERS
AND SAMPLING ALGORITHMS

Algorithm	Parameters
MLP	training: rprop; iterations $\in [100, 200, \dots, 1000]$; #hidden neurons = $\frac{\text{\#input} + \text{\#output}}{2}$
k-NN	nearest neighbors $\in [1, 3, \dots, 11]$
MC-RBO	$\gamma \in [0.001, 0.01, \dots, 10]$; step $\in [0.001, 0.005, \dots, 0.1]$; iterations $\in [1000, 2000, \dots, 8000]$; oversampling ratio $\in [50, 100, \dots, 500]$; neighbors for potential approximations $\in [10, 25, 50, 100]$
SMOTE-all	k-nearest neighbors = 5; oversampling ratio $\in [50, 100, \dots, 500]$
S-SMOTE [31]	k-nearest neighbors = 5; oversampling ratio $\in [50, 100, \dots, 500]$
MDO [35]	$K1 \in [1, 2, \dots, 10]$; $K2 \in [2, 4, \dots, 20]$; oversampling ratio $\in [50, 100, \dots, 500]$
SMOM [32]	$K1 \in [2, 4, \dots, 20]$; $K2 \in [1, 2, \dots, 10]$; $rTh \in [0.1, 0.2, \dots, 1]$; $rTh \in [1, 2, \dots, 10]$; $w1, w2, r1, r2 \in [0.1, 0.2, \dots, 1]$; k-nearest neighbors = 5; oversampling ratio $\in [50, 100, \dots, 500]$
NB-SVM [44]	kernel = RBF; $C \in [1.0, 1.5, \dots, 5.0]$; $\gamma \in [0.001, 0.002, \dots, 0.01]$; training = modified SMO $\sigma \in [1, 2, 3, 4, 5, 10, 20, \dots, 50]$ OVA = max of supports
AdaB.NC [8]	no. of iterations $\in [5, 10, \dots, 50]$ $\lambda \in [0, 1, \dots, 4]$
SMOTEBag [45]	no. of classifiers $\in [5, 10, \dots, 50]$ no. of neighbors for SMOTE = 5; oversampling ratio = 100%
Binarization RUS	one-vs-one combination = weighted voting undersampling ratio $\in [10, 20, \dots, 90]$
SMOTE [5]	k-nearest neighbors = 5; oversampling ratio $\in [50, 100, \dots, 500]$
BSMOTE [14]	k-nearest neighbors = 5; oversampling ratio $\in [50, 100, \dots, 500]$
ADASYN [46]	k-nearest neighbors = 5; $\beta \in [0.1, 0.2, \dots, 1]$

3) *Classifier Validation Procedure*: For training and testing of examined classifiers, we have employed a stratified tenfold cross validation, repeated ten times. All parameters for each tested algorithm were established for each data set independently using an internal stratified threefold cross validation on the training set.

4) *Performance Metrics*: We evaluate the performance of classifiers on multiclass imbalanced data using four dedicated skew-insensitive metrics [47]: average accuracy (AvAcc), class balance accuracy (CBA), multiclass G-measure (mGM), and confusion entropy (CEN). They are expressed as follows:

$$\text{AvAcc} = \frac{\sum_{i=1}^M TPR_i}{M} \quad (2)$$

$$\text{CBA} = \frac{\sum_{i=1}^M \frac{\text{mat}_{i,i}}{\max(\sum_{j=1}^M \text{mat}_{i,j}, \sum_{j=1}^M \text{mat}_{j,i})}}{M} \quad (3)$$

$$\text{mGM} = \sqrt[M]{\prod_{i=1}^M \text{recall}_i} \quad (4)$$

$$\text{CEN} = \sum_{i=1}^M P_i \cdot \text{CEN}_i \quad (5)$$

TABLE III

SHAFFER'S TESTS FOR COMPARISON BETWEEN MC-RBO_{OVA+} AND REFERENCE MC-RBO_{OVO} AND MC-RBO_{OVA} FOR FOUR USED BASE CLASSIFIER. WE REPORT OBTAINED *p*-VALUES. SYMBOL ">" STANDS FOR SITUATION IN WHICH MC-RBO_{OVA+} IS STATISTICALLY SUPERIOR AND "=" FOR WHEN THERE ARE NO SIGNIFICANT DIFFERENCES

Classifier	Hypothesis	AvACC	CBA	mGM	CEN
C5.0	vs. MC-RBO _{OVO}	> (0.0367)	> (0.0146)	> (0.0389)	> (0.0104)
	vs. MC-RBO _{OVA}	> (0.0393)	> (0.0288)	> (0.0371)	> (0.0379)
MLP	vs. MC-RBO _{OVO}	= (0.1782)	> (0.0399)	= (0.2088)	= (0.0931)
	vs. MC-RBO _{OVA}	= (0.1290)	> (0.0207)	= (0.1872)	= (0.0633)
k-NN	vs. MC-RBO _{OVO}	> (0.0218)	> (0.0211)	= (0.0768)	> (0.0294)
	vs. MC-RBO _{OVA}	= (0.305)	> (0.0278)	= (0.1003)	> (0.0416)
NB	vs. MC-RBO _{OVO}	> (0.0321)	> (0.0116)	> (0.0368)	> (0.0096)
	vs. MC-RBO _{OVA}	> (0.0335)	> (0.0229)	> (0.0400)	> (0.0154)

where M is the number of classes, $\text{mat}_{i,j}$ stands for the number of instances of the true class i , which were predicted as class j , $P_i = (\sum_{j=1}^M \text{mat}_{i,j} + \text{mat}_{j,i} / 2 \cdot \sum_{l=1}^M \text{mat}_{k,l})$, and $\text{CEN}_i = -\sum_{j=1, i \neq j}^M (P_{i,j} \cdot \log_{2(C-1)}(P_{i,j}) + P_{j,i} \cdot \log_{2(C-1)}(P_{j,i}))$. In addition, for CEN, we have $P_{i,i} = 0$, $P_{i,j} = \text{mat}_{i,j} / (\sum_{j=1}^M \text{mat}_{i,j} + \text{mat}_{j,i})$, and $i \neq j$.

5) *Statistical Analysis of Results*: To offer a more detailed comparison among methods, we have conducted various types of statistical tests of significance. For pairwise comparison over a single data set, we have employed a combined tenfold cross-validation F-test [48]. For comparison of methods over multiple data sets, we have used a Friedman ranking test with Shaffer *post hoc* analysis [49]. All employed tests of significance used confidence level $\alpha = 0.05$.

C. Results and Discussion

First, we wanted to analyze the role of sampling techniques applied to calculate the MC-RBO potential functions from $M-1$ nontarget classes (i.e., other than the one currently being oversampled). We compare the proposed approach in which we select a random subset of instances from each class (named OVA+) by taking into account all of the instances (OVA) and calculating potential function values in a pairwise manner and then averaging it (OVO). Results averaged over 20 data sets for four metrics and four base classifiers are shown in Fig. 5, while the outputs of *post hoc* statistical analysis are depicted in Table III.

Due to space restrictions, the detailed comparison with reference methods for each data set is presented for C5.0 classifier only in Fig. 6 and the details can be found in Tables IV–VII. Outputs of pairwise statistical tests for all four classifiers and four metrics are presented in Fig. 7, while Tables VIII–XI depict the outcomes of *post hoc* statistical tests. The role of varying oversampling ration on the performance of multiclass sampling methods is shown in Fig. 7. As trends in the influence of oversampling ratios on achieved performance were similar for all of the examined classifiers, we decided to present plots for C5.0 as representative ones.

1) *Role of Class Sampling in MC-RBO Potential Function Calculation*: Obtained results clearly prove the superiority of the proposed sampling approach for calculating class potentials (OVA+). By selecting only a subset of original examples, we can maintain the characteristics of each class while achieving a speedup of the calculation process, as well as reduce the

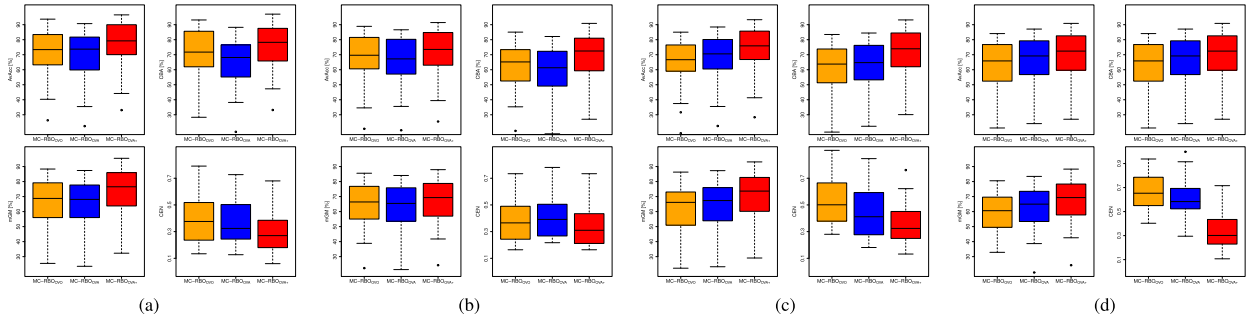


Fig. 5. Comparison of different sampling approaches for MC-RBO with respect to performance measures and base classifiers, averaged over 20 data sets. (a) C5.0. (b) MLP. (c) k -NN. (d) NB.

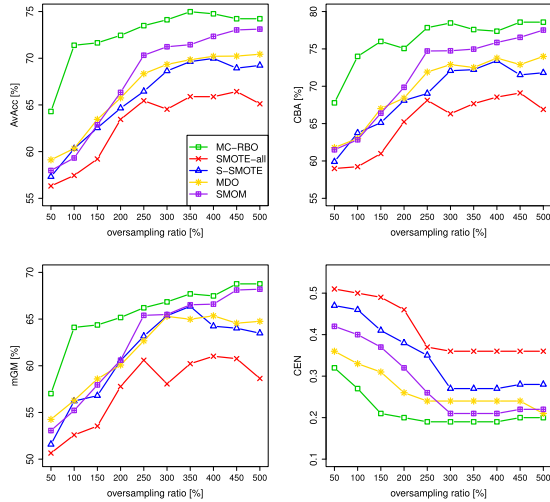


Fig. 6. Effects of the varying oversampling ratio on the performance of examined multiclass oversampling algorithms with C5.0 classifier, averaged over 20 data sets.

dominance of majority classes during the potential function calculation. The standard approach that takes into account all instances from all classes during potential calculation (OVA) is too slow and may be easily destabilized by outliers or strong class imbalance. On the other hand, calculating pairwise potentials between classes and then averaging them (OVO) returns highly unsatisfactory performance. This can be contributed to the lack of correlations between pairwise and nonpairwise class characteristics.

2) *Impact of Varying Oversampling Ratios:* Fig. 6 depicts how oversampling ratios affect the performance of five evaluated multiclass oversampling algorithms. The most interesting observation lies in the fact that the proposed MC-RBO offers significantly better performance for small oversampling ratios. While other methods tend to improve their performance when the oversampling ratio is higher than 200%, MC-RBO starts giving a stable performance for ratios of 100%. This shows that MC-RBO does not need to introduce as many artificial instances as reference algorithms, as they are introduced in a much more guided manner. Therefore, we can conclude that MC-RBO relies not on the number of created instances, but on the intelligent procedure that places them in important parts of the decision space, offering significantly better empowering of minority classes.

3) *Impact of Base Classifier Selection:* Interestingly, MC-RBO on average performs well with all of the examined

classifiers. This is a desirable property, as sampling methods should be treated as filter approaches—they should work well without targeting a specific learning model. By more detailed analysis of results, one can see that MC-RBO especially favors C5.0 and NB classifiers. For C5.0, this can be explained by a more guided introduction of new instances by MC-RBO. As the new instances do not cause data set shift or increased learning difficulties on class borders, C5.0 can conduct more efficient split search and do not overfit to data. NB benefits from MC-RBO as well, as new instances generated by our algorithm follow the underlying density distributions of each class and thus allow for efficient estimation of NB probabilities.

MC-RBO performs slightly worse for MLP and k -NN, which can be explained by the nature of these classifiers. Both of them benefit from a higher number of training instances, allowing for better capturing learned concepts. As MC-RBO usually generates fewer instances than reference methods, these two classifiers may display some drop in performance. This could be potentially alleviated by forcing MC-RBO to generate more instances when working with such base classifiers.

4) *Comparison With Multiclass Sampling Approaches:* MC-RBO returned superior performance to all of multiclass sampling methods, regardless of the used base classifier or performance metric. All four reference methods utilized different approaches to create artificial instances (using NNs or Mahalanobis distance), but they all had one thing in common—all of them used only information coming from the class currently being oversampled. In multiclass scenarios, such an approach can be seen as an oversimplification and may lead to introducing artificial instances on borders between multiple classes or even increasing the actual overlapping among classes. As MC-RBO utilized information coming from all of the classes, it was able to create a much better set of artificial instances, which directly translated into improved classification rates.

5) *Comparison With Multiclass Algorithm-Level Approaches:* When being compared with HTree and NB-SVM, the proposed MC-RBO returns significantly better performance in almost all of the cases. This shows that while algorithm-level approaches are effective for binary problems, they have a long way ahead to secure similar status in multiclass imbalanced domains. The relationships among classes are too complex to be effectively modeled by simple

TABLE IV
RESULTS ACCORDING TO AVACC [%] METRIC FOR MC-RBO AND REFERENCE METHODS WITH C5.0 AS BASE CLASSIFIER

	Multi-class sampling approaches				Inbuilt mechanisms				Binarization+sampling				
Dataset	MC-RBO	SMOTE-all	S-SMOTE	MDO	SMOM	Htree	NB-SVM	AdaB.NC	SMOTEBag	RUS	SMOTE	BSMOTE	ADASYN
Automobile	77.11	80.12	73.53	78.13	79.04	62.97	70.73	75.16	69.85	60.56	78.94	80.86	76.06
Balance	85.32	55.06	55.01	57.70	59.52	82.90	85.81	63.58	58.37	54.49	56.99	57.95	55.07
Car	96.43	89.84	90.13	93.36	95.18	49.01	54.83	91.10	94.49	94.21	91.93	91.93	93.85
Cleveland	35.01	28.92	27.18	28.92	28.01	25.29	31.11	27.61	32.28	31.92	30.07	30.07	26.23
Contraceptive	51.93	50.63	46.92	53.27	55.09	44.30	47.21	50.14	49.28	48.59	48.76	48.76	45.88
Dermatology	96.11	95.72	96.10	97.48	99.31	87.72	96.45	94.44	94.68	94.13	95.37	93.45	94.41
Ecoli	72.06	64.68	67.54	61.16	61.16	55.06	60.88	65.03	63.48	58.68	66.78	68.70	68.70
Flare	70.63	63.73	66.27	61.09	62.19	45.72	54.45	60.58	64.72	63.81	61.90	60.94	58.06
Glass	71.86	71.52	68.72	70.64	71.55	57.11	61.96	73.10	72.79	64.11	68.86	66.94	67.90
Hayes-roth	91.88	86.45	88.04	87.33	90.06	70.60	76.42	87.56	86.28	86.18	86.60	84.68	83.72
Led7digit	71.53	72.39	72.55	75.03	75.94	53.63	59.45	22.39	70.91	70.03	70.34	71.30	68.42
Lymphography	81.22	73.02	62.67	76.54	74.72	65.98	69.86	69.57	65.29	48.43	70.16	74.00	66.32
Newthyroid	95.76	94.70	93.48	92.06	90.24	87.11	89.05	92.70	93.85	92.87	90.47	94.31	93.35
Pageblocks	83.71	75.83	75.25	78.47	77.56	39.27	48.97	66.58	85.19	82.08	86.28	90.12	89.16
Thyroid	89.43	80.02	85.34	79.14	80.96	58.18	64.97	90.82	92.90	87.54	90.02	90.05	89.06
Vehicle	79.48	73.49	73.71	70.85	70.85	63.07	69.86	76.08	75.99	69.14	70.19	74.03	73.07
Wine	97.57	92.53	90.80	93.41	93.41	88.26	96.02	94.56	95.88	91.26	91.93	90.97	90.97
Winequality-red	44.12	37.41	35.79	40.05	42.78	27.96	31.84	37.71	38.97	33.46	34.51	33.55	31.63
Yeast	60.39	51.03	52.42	54.55	56.37	44.52	52.28	57.77	51.39	49.92	47.77	47.77	47.77
Zoo	88.56	82.61	68.69	79.09	79.09	82.86	87.62	90.83	70.20	66.92	72.81	76.65	75.69
Avg. rank	2.70	6.32	7.65	5.88	5.00	11.80	8.70	6.10	5.55	9.20	7.18	6.42	8.50

TABLE V
RESULTS ACCORDING TO CBA [%] METRIC FOR MC-RBO AND REFERENCE METHODS WITH C5.0 AS BASE CLASSIFIER

	Multi-class sampling approaches					Inbuilt mechanisms				Binarization+sampling			
Dataset	MC-RBO	SMOTE-all	S-SMOTE	MDO	SMOM	Htree	NB-SVM	AdaB.NC	SMOTEBag	RUS	SMOTE	BSMOTE	ADASYN
Automobile	80.48	54.47	71.79	75.11	73.35	57.75	69.86	58.25	68.94	57.08	74.59	74.77	66.49
Balance	69.12	45.79	55.88	57.70	60.34	61.16	77.98	52.95	56.55	46.66	50.03	51.86	48.11
Car	96.53	87.25	89.26	95.73	98.37	43.79	53.09	94.21	95.40	88.99	88.45	84.97	88.63
Cleveland	35.01	24.09	25.44	27.34	27.34	18.33	25.89	30.38	30.46	25.83	26.59	25.72	16.66
Contraceptive	52.77	41.63	44.31	51.69	52.97	42.56	45.47	50.90	45.64	42.50	43.54	40.93	41.53
Dermatology	94.22	86.30	94.36	95.64	93.88	82.50	91.23	91.82	93.77	88.91	90.15	89.10	84.84
Ecoli	72.01	56.07	68.41	63.53	61.77	52.45	58.27	60.22	60.75	50.85	63.30	65.22	60.87
Flare	63.56	58.59	61.92	59.51	61.27	43.98	52.71	63.04	64.72	59.46	56.68	53.98	51.10
Glass	75.03	60.63	66.11	70.64	69.76	51.89	61.09	64.88	70.97	60.63	61.90	63.46	60.07
Hayes-roth	88.64	77.48	86.30	84.96	87.62	63.64	72.07	87.72	82.64	78.35	83.12	76.85	77.63
Led7digit	75.12	63.94	70.81	78.19	79.95	46.67	54.23	67.72	67.27	65.68	65.12	65.21	64.07
Lymphography	79.02	43.21	59.19	75.75	78.39	59.02	64.64	49.97	68.02	44.95	64.07	68.78	59.36
Newthyroid	92.84	85.04	90.00	95.22	93.46	80.15	85.57	92.87	93.85	87.65	84.38	89.09	84.65
Pageblocks	86.73	78.60	71.77	80.84	79.96	35.79	46.36	82.85	81.55	76.86	81.06	83.16	82.20
Thyroid	84.11	80.58	81.86	76.77	75.01	55.57	57.14	88.31	93.87	80.58	85.67	85.07	81.23
Vehicle	78.05	62.18	70.23	72.43	72.43	58.72	62.90	69.14	78.72	62.18	64.10	67.07	65.24
Wine	94.01	84.30	91.67	91.83	90.95	81.30	93.41	93.57	92.24	87.78	87.58	83.14	83.14
Winequality-red	47.66	24.76	34.92	41.63	39.87	23.61	30.10	31.15	40.79	27.37	31.03	29.20	24.67
Yeast	57.97	46.44	48.94	53.76	55.52	42.78	44.45	48.38	47.75	42.96	41.68	42.55	39.94
Zoo	82.51	60.83	66.08	76.72	76.72	76.77	81.53	65.38	72.02	63.44	68.46	72.30	66.99
Avg. rank	1.70	10.62	6.00	3.88	4.22	11.30	8.15	5.55	4.40	9.68	7.80	7.77	9.93

TABLE VI
RESULTS ACCORDING TO MGM [%] METRIC FOR MC-RBO AND REFERENCE METHODS WITH C5.0 AS BASE CLASSIFIER

	Multi-class sampling approaches					Inbuilt mechanisms				Binarization+sampling			
Dataset	MC-RBO	SMOTE-all	S-SMOTE	MDO	SMOM	Htree	NB-SVM	AdaB.NC	SMOTEBag	RUS	SMOTE	BSMOTE	ADASYN
Automobile	77.20	51.86	74.40	75.11	78.75	56.88	64.64	57.38	69.81	55.54	73.72	74.00	62.14
Balance	61.93	40.57	52.40	55.92	58.65	59.42	77.11	55.56	53.94	47.43	47.42	50.32	50.72
Car	97.15	81.16	91.00	95.73	94.82	42.92	53.09	95.95	96.27	88.22	84.10	82.66	87.76
Cleveland	34.04	18.00	24.57	26.45	24.63	19.20	23.28	30.38	29.59	24.29	26.59	21.26	14.92
Contraceptive	50.99	38.15	43.44	51.69	52.60	42.56	41.12	52.64	44.77	43.27	42.67	40.93	37.18
Dermatology	95.72	79.34	95.23	92.08	92.08	80.76	86.01	91.82	94.64	88.14	85.80	89.64	84.84
Ecoli	72.66	51.72	67.54	63.53	63.53	52.45	53.92	62.83	59.01	51.62	62.43	61.22	59.13
Flare	65.50	55.98	61.92	58.62	60.44	44.85	50.10	63.04	60.37	59.46	56.68	56.29	51.10
Glass	73.08	55.41	62.63	67.97	67.97	51.89	55.00	66.62	71.84	59.86	57.55	60.46	59.20
Hayes-roth	81.93	70.52	84.56	83.18	86.82	62.77	66.85	89.46	78.29	77.58	81.38	77.62	77.63
Led7digit	75.02	57.85	70.81	78.19	77.28	47.54	48.14	71.20	66.40	63.37	65.99	63.67	65.81
Lymphography	79.13	40.60	58.32	73.08	73.99	59.89	65.51	49.10	68.89	44.95	62.33	67.24	55.01
Newthyroid	96.73	80.69	86.52	92.55	93.46	79.28	84.70	95.48	93.85	85.34	80.90	87.55	83.78
Pageblocks	87.27	71.64	74.38	79.06	77.24	34.05	46.36	84.59	79.81	76.86	80.19	84.70	77.85
Thyroid	86.94	73.62	81.86	75.88	77.70	55.57	56.27	91.79	93.87	78.27	85.67	80.61	78.62
Vehicle	70.47	59.57	71.97	70.65	70.65	57.85	58.55	70.88	76.98	63.23	62.36	69.38	66.11
Wine	91.73	80.82	93.41	90.05	89.14	79.56	87.32	92.70	88.76	87.01	84.10	80.83	78.79
Winequality-red	48.33	17.80	36.66	40.74	42.56	21.87	30.10	32.89	41.66	27.37	28.42	30.74	22.93
Yeast	58.06	43.83	52.42	51.09	52.91	43.65	40.10	51.86	45.14	41.42	42.55	43.32	42.55
Zoo	85.01	55.61	63.47	75.83	77.65	76.77	82.40	64.51	72.02	64.21	67.59	72.30	66.12
Avg. rank	2.35	11.55	5.30	4.45	3.85	10.90	9.35	4.30	4.50	9.35	7.87	7.60	9.63

modifications of training functions that have roots in two-class problems. On the other hand, AdaB.NC and SMOTEBag were much more formidable counterparts for MC-RBO, being

able of returning statistically significantly better performance than MC-RBO on few data sets. Remember that both of these algorithms are, in fact, ensemble approaches utilizing

TABLE VII
RESULTS ACCORDING TO CEN METRIC FOR MC-RBO AND REFERENCE METHODS WITH C5.0 AS BASE CLASSIFIER

Dataset	Multi-class sampling approaches						Inbuilt mechanisms			Binarization+sampling			
	MC-RBO	SMOTE-all	S-SMOTE	MDO	SMOM	Htree	NB-SVM	AdaB.NC	SMOTEBag	RUS	SMOTE	BSMOTE	ADASYN
Automobile	0.23	0.56	0.29	0.29	0.30	0.50	0.53	0.49	0.31	0.46	0.48	0.30	0.38
Balance	0.37	0.64	0.49	0.48	0.46	0.43	0.57	0.50	0.47	0.55	0.55	0.51	0.51
Car	0.10	0.26	0.11	0.09	0.04	0.64	0.66	0.09	0.06	0.19	0.19	0.21	0.12
Cleveland	0.65	0.87	0.76	0.75	0.76	0.89	0.93	0.76	0.73	0.83	0.94	0.80	0.86
Contraceptive	0.50	0.69	0.59	0.49	0.44	0.65	0.72	0.51	0.57	0.60	0.72	0.60	0.63
Dermatology	0.04	0.27	0.06	0.13	0.14	0.26	0.31	0.13	0.08	0.14	0.20	0.11	0.15
Ecoli	0.30	0.53	0.34	0.41	0.40	0.52	0.59	0.43	0.45	0.54	0.51	0.41	0.41
Flare	0.36	0.51	0.39	0.46	0.41	0.58	0.62	0.40	0.44	0.43	0.54	0.45	0.51
Glass	0.36	0.53	0.38	0.34	0.34	0.54	0.64	0.40	0.32	0.45	0.48	0.41	0.42
Hayes-roth	0.19	0.34	0.19	0.19	0.14	0.40	0.47	0.17	0.23	0.25	0.27	0.23	0.24
Led7digit	0.16	0.51	0.32	0.24	0.21	0.56	0.64	0.35	0.37	0.43	0.39	0.37	0.36
Lymphography	0.26	0.68	0.43	0.31	0.34	0.43	0.49	0.55	0.35	0.61	0.52	0.35	0.46
Newthyroid	0.06	0.28	0.14	0.08	0.12	0.27	0.28	0.08	0.08	0.21	0.24	0.14	0.16
Pageblocks	0.15	0.34	0.29	0.24	0.19	0.72	0.70	0.18	0.23	0.27	0.29	0.19	0.23
Thyroid	0.09	0.34	0.22	0.27	0.29	0.50	0.59	0.12	0.09	0.24	0.28	0.21	0.23
Vehicle	0.28	0.47	0.30	0.34	0.33	0.46	0.54	0.36	0.29	0.43	0.40	0.33	0.35
Wine	0.07	0.25	0.10	0.15	0.18	0.25	0.24	0.13	0.13	0.18	0.26	0.23	0.21
Winequality-red	0.53	0.86	0.67	0.64	0.59	0.83	0.88	0.73	0.60	0.77	0.87	0.71	0.79
Yeast	0.42	0.62	0.51	0.54	0.47	0.59	0.73	0.54	0.56	0.65	0.67	0.61	0.59
Zoo	0.21	0.48	0.38	0.27	0.27	0.27	0.34	0.41	0.32	0.43	0.40	0.32	0.35
Avg. rank	1.45	11.42	4.65	4.27	3.93	10.00	12.10	5.52	4.40	8.93	10.35	6.32	7.65

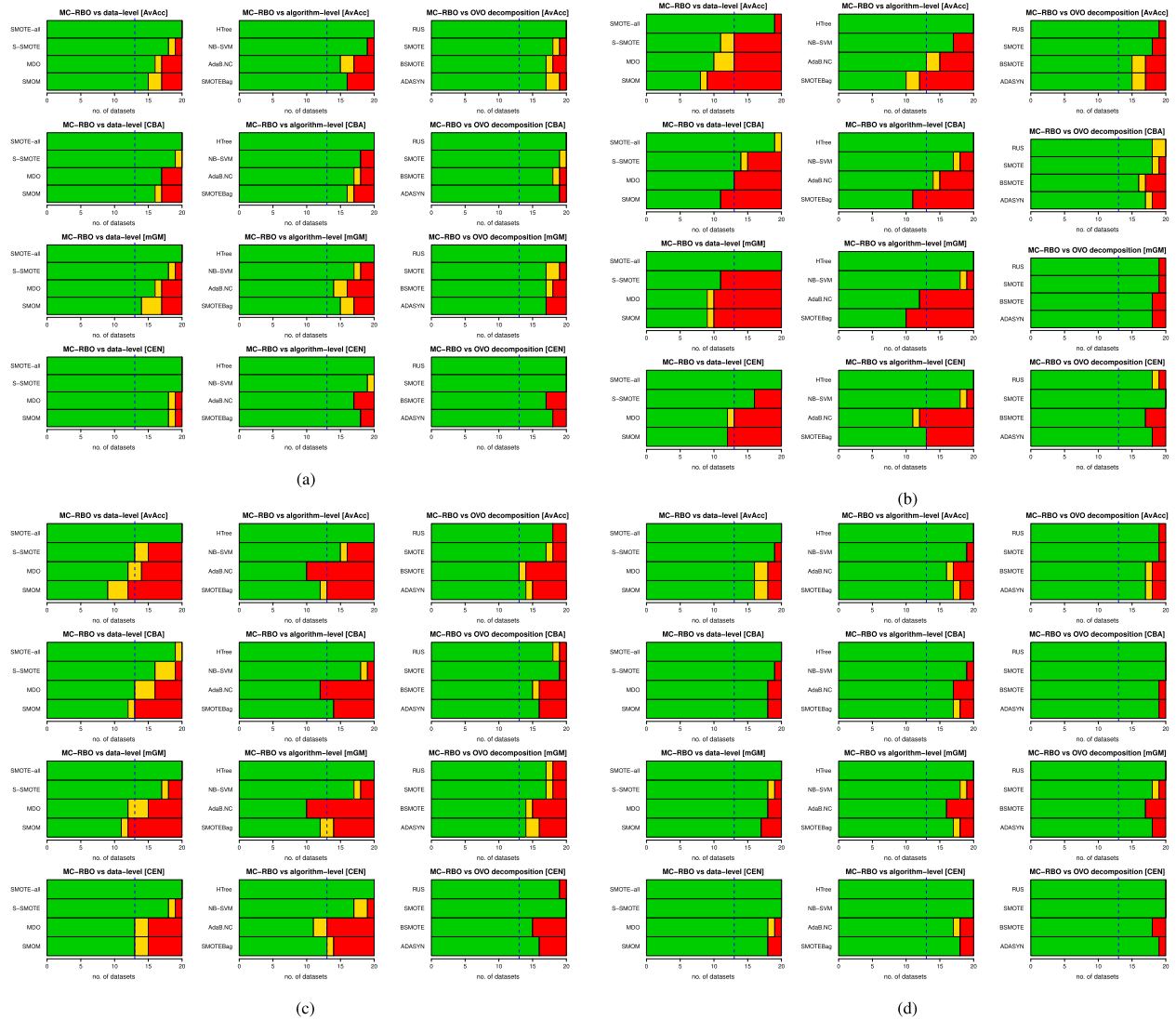


Fig. 7. Comparison of MC-RBO with reference methods for four tested base classifiers with respect to the number of data sets on which MC-RBO was statistically significantly better (green), similar (yellow), or worse (red) using combined tenfold CV F-test over 20 data sets. Dashed lines: critical values with a confidence level $\alpha = 0.05$ for a *post hoc* analysis. (a) C5.0. (b) MLP. (c) k -NN. (d) NB.

multiple classifiers. Therefore, in fact, we were comparing a single classifier (MC-RBO) with ensembles. This further shows the quality of MC-RBO, as it was able to behave better than combined classifiers on a plethora of multiclass

TABLE VIII

SHAFFER'S TESTS FOR COMPARISON BETWEEN MC-RBO AND REFERENCE METHODS FOR C5.0 CLASSIFIER. WE REPORT OBTAINED p -VALUES. SYMBOL ">" STANDS FOR SITUATION IN WHICH MC-RBO IS STATISTICALLY SUPERIOR AND "=" FOR WHEN THERE ARE NO SIGNIFICANT DIFFERENCES

Hypothesis	AvACC	CBA	mGM	CEN
vs. SMOTE-all	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. S-SMOTE	> (0.0069)	> (0.0000)	> (0.0087)	> (0.0000)
vs. MDO	> (0.0325)	> (0.0203)	> (0.0341)	> (0.0069)
vs. SMOM	> (0.0374)	> (0.0248)	> (0.0409)	> (0.0073)
vs. HTree	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. NB-SVM	> (0.0053)	> (0.0122)	> (0.0151)	> (0.0027)
vs. AdaB.NC	> (0.0401)	> (0.1988)	> (0.0389)	> (0.0179)
vs. SMOTEBag	> (0.0411)	> (0.0272)	> (0.0455)	> (0.0139)
vs. OVO-RUS	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. OVO-SMOTE	> (0.0102)	> (0.0020)	> (0.0214)	> (0.0000)
vs. OVO-BSMOTE	> (0.0218)	> (0.0125)	> (0.0188)	> (0.0233)
vs. OVO-ADASYN	> (0.0193)	> (0.0089)	> (0.0211)	> (0.0181)

TABLE IX

SHAFFER'S TESTS FOR COMPARISON BETWEEN MC-RBO AND REFERENCE METHODS FOR MLP CLASSIFIER. WE REPORT OBTAINED p -VALUES. SYMBOL ">" STANDS FOR SITUATION IN WHICH MC-RBO IS STATISTICALLY SUPERIOR AND "=" FOR WHEN THERE ARE NO SIGNIFICANT DIFFERENCES

Hypothesis	AvACC	CBA	mGM	CEN
vs. SMOTE-all	> (0.0082)	> (0.0031)	> (0.0000)	> (0.0000)
vs. S-SMOTE	= (0.2481)	> (0.0441)	= (0.3982)	> (0.0386)
vs. MDO	= (0.3319)	> (0.0498)	= (0.9821)	= (0.2977)
vs. SMOM	< (0.0432)	= (0.4098)	= (0.9821)	= (0.0899)
vs. HTree	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. NB-SVM	> (0.0273)	> (0.0208)	> (0.0174)	> (0.0174)
vs. AdaB.NC	> (0.0439)	> (0.0385)	= (0.0871)	= (0.1184)
vs. SMOTEBag	= (0.2066)	= (0.3590)	= (0.5824)	= (0.0416)
vs. OVO-RUS	> (0.0048)	> (0.0017)	> (0.0048)	> (0.0052)
vs. OVO-SMOTE	> (0.0092)	> (0.0071)	> (0.0048)	> (0.0000)
vs. OVO-BSMOTE	> (0.0285)	> (0.0327)	> (0.0195)	> (0.0303)
vs. OVO-ADASYN	> (0.0285)	> (0.0319)	> (0.0195)	> (0.0195)

TABLE X

SHAFFER'S TESTS FOR COMPARISON BETWEEN MC-RBO AND REFERENCE METHODS FOR k -NN CLASSIFIER. WE REPORT OBTAINED p -VALUES. SYMBOL ">" STANDS FOR SITUATION IN WHICH MC-RBO IS STATISTICALLY SUPERIOR AND "=" FOR WHEN THERE ARE NO SIGNIFICANT DIFFERENCES

Hypothesis	AvACC	CBA	mGM	CEN
vs. SMOTE-all	> (0.0000)	> (0.0014)	> (0.0000)	> (0.0000)
vs. S-SMOTE	> (0.0441)	> (0.0146)	> (0.0202)	> (0.0131)
vs. MDO	= (0.0955)	> (0.0401)	= (0.0515)	> (0.0451)
vs. SMOM	= (0.4936)	= (0.0996)	= (0.1219)	> (0.0418)
vs. HTree	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. NB-SVM	> (0.0321)	> (0.0088)	> (0.0135)	> (0.0096)
vs. AdaB.NC	= (0.5228)	= (0.1998)	> (0.5688)	= (0.0633)
vs. SMOTEBag	= (0.0587)	> (0.0399)	> (0.0455)	> (0.0373)
vs. OVO-RUS	> (0.0121)	> (0.0103)	> (0.0111)	> (0.0078)
vs. OVO-SMOTE	> (0.0171)	> (0.0096)	> (0.0129)	> (0.0000)
vs. OVO-BSMOTE	> (0.0387)	> (0.0301)	> (0.0363)	> (0.0351)
vs. OVO-ADASYN	> (0.0364)	> (0.0288)	> (0.0366)	> (0.0307)

data. Finally, MC-RBO can be easily combined with any ensemble learning algorithm, which should further increase its predictive power. We will explore this direction in future works.

6) *Comparison With Binary Decomposition Approaches:* Finally, the comparison of MC-RBO with OVO solutions shows the superiority of using *ad hoc* methodologies. Binary decomposition approaches returned inferior performance to most data sampling and algorithm-level solutions. Despite using advanced binary sampling techniques (such as BSMOTE or ADASYN), treating multiclass problems as

TABLE XI

SHAFFER'S TESTS FOR COMPARISON BETWEEN MC-RBO AND REFERENCE METHODS FOR NB CLASSIFIER. WE REPORT OBTAINED p -VALUES. SYMBOL ">" STANDS FOR SITUATION IN WHICH MC-RBO IS STATISTICALLY SUPERIOR AND "=" FOR WHEN THERE ARE NO SIGNIFICANT DIFFERENCES

Hypothesis	AvACC	CBA	mGM	CEN
vs. SMOTE-all	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. S-SMOTE	> (0.0047)	> (0.0000)	> (0.0067)	> (0.0000)
vs. MDO	> (0.0262)	> (0.0172)	> (0.0298)	> (0.0039)
vs. SMOM	> (0.0391)	> (0.0200)	> (0.0370)	> (0.0089)
vs. HTree	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. NB-SVM	> (0.0064)	> (0.0097)	> (0.0149)	> (0.0024)
vs. AdaB.NC	> (0.0328)	> (0.0148)	> (0.0360)	> (0.0203)
vs. SMOTEBag	> (0.0386)	> (0.0194)	> (0.0445)	> (0.0080)
vs. OVO-RUS	> (0.0000)	> (0.0000)	> (0.0000)	> (0.0000)
vs. OVO-SMOTE	> (0.0051)	> (0.0000)	> (0.0147)	> (0.0000)
vs. OVO-BSMOTE	> (0.0166)	> (0.0049)	> (0.0202)	> (0.0188)
vs. OVO-ADASYN	> (0.0201)	> (0.0041)	> (0.0248)	> (0.0106)

pairwise ones is not a good solution. Many data-level phenomena will be omitted by such a decomposition, including cases in which more than two classes overlap or when the decision boundary has in its vicinity instances with similar characteristics, but originating from multiple classes. Furthermore, detecting outliers or small disjuncts is not an easy task when using binary decomposition, as we no longer maintain the complex view on underlying class structures. This allows us to conclude that developing native multiclass solutions is a much more efficient direction than using binary decomposition, and MC-RBO was developed with this goal in mind.

V. CONCLUSION

The main aim of this paper was to propose a novel, effective oversampling approach for a challenging task of multiclass imbalanced data classification. We proposed the MC-RBO algorithm, which can exploit local data characteristics of each class and offers better placement of new artificially generated instances and more targeted empowering of minority classes. As distinct from SMOTE-like algorithms, MC-RBO does not use NNs of minority instances, and thus, it is robust to atypical data distributions, especially when the minority classes form several disjoint clusters. The computer experiments confirmed the usefulness of the proposed approach and on the basis of a thorough statistical analysis we may assert that for many types of data sets MC-RBO is statistically significantly better than state-of-art methods. Particularly, it outperforms existing multiclass oversampling methods, as well as typical binary decomposition scenarios.

This paper is a step forward toward the use of non-NN-based oversampling for multiclass imbalanced data. Results obtained in this paper encourage us to continue works on alternative approaches to oversampling imbalanced data, with a special focus on the following issues.

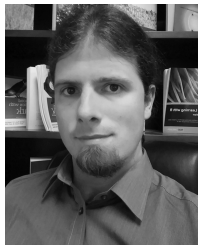
- 1) Evaluate how MC-RBO is robust to class label and feature noise.
- 2) Evaluate if using different similarity metrics, especially fractional for high-dimensional data [50], can improve the quality of MC-RBO.
- 3) Propose novel metrics based on radial functions of imbalance distribution, which can improve the algorithms, especially points out where the artificially generated examples should be placed.

- 4) Propose how to use MC-RBO in incremental (or block) mode for imbalanced data stream classification.
- 5) Embed MC-RBO into hybrid architectures with inbuilt mechanisms.

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